

Recurrence Tree Explanation



$$T(n) = 3T(n/4) + O(n^2)$$

explain me the recurrence tree solve for this in detailed explanation

To solve the recurrence

$$T(n) = 3T(n/4) + O(n^2)$$

using the **recurrence tree method**, we'll break it down step by step, visualizing how the cost accumulates at each level.

♦ Step 1: Understand the Structure of the Recurrence

This recurrence has:

- **Number of subproblems:** 3
- **Size of each subproblem:** $\frac{n}{4}$
- **Cost at each level (outside recursion):** $O(n^2)$

This structure indicates **divide and conquer**.

♦ Step 2: Build the Recurrence Tree

We start expanding the recurrence to see how the work is distributed at each level.

■ Level 0 (Root):

- Work done: $O(n^2)$
- Breaks into: 3 subproblems of size $\frac{n}{4}$

■ Level 1:

- Each of the 3 subproblems does $O((n/4)^2) = O(n^2/16)$ work.
- Total work:

$$3 \cdot O(n^2/16) = O\left(\frac{3n^2}{16}\right)$$

Level 2:

- Each of the 3 subproblems at level 1 creates 3 more subproblems of size $\frac{n}{4^2} = \frac{n}{16}$
- Number of subproblems: $3^2 = 9$
- Work per subproblem: $O((n/16)^2) = O(n^2/256)$
- Total work:

$$9 \cdot O(n^2/256) = O\left(\frac{9n^2}{256}\right)$$

Level k (in general):

- Number of subproblems: 3^k
- Size of each subproblem: $\frac{n}{4^k}$
- Work per subproblem:

$$O\left(\left(\frac{n}{4^k}\right)^2\right) = O\left(\frac{n^2}{16^k}\right)$$

- Total work at level k :

$$3^k \cdot \frac{n^2}{16^k} = n^2 \cdot \left(\frac{3}{16}\right)^k$$

Step 3: Add Work Across All Levels

$$T(n) = \sum_{k=0}^{\log_4 n} n^2 \cdot \left(\frac{3}{16}\right)^k$$

This is a **geometric series** with ratio $r = \frac{3}{16} < 1$, so it **converges**.

$$T(n) = n^2 \cdot \sum_{k=0}^{\log_4 n} \left(\frac{3}{16}\right)^k \leq n^2 \cdot \sum_{k=0}^{\infty} \left(\frac{3}{16}\right)^k$$

The sum of an infinite geometric series is:

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r} = \frac{1}{1-3/16} = \frac{16}{13}$$

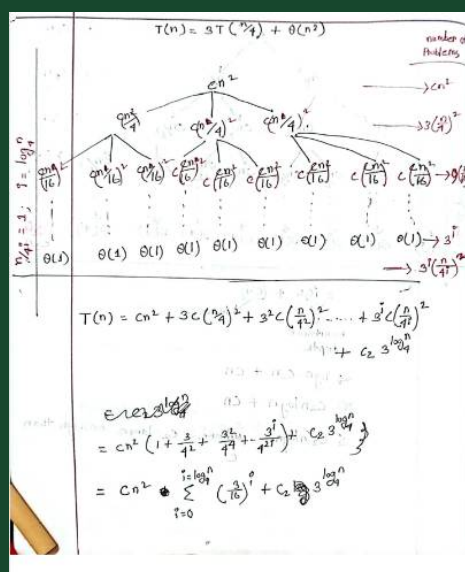
So the total work is:

$$T(n) = O(n^2)$$

Final Answer:

$$T(n) = O(n^2)$$

This means the **total time complexity is dominated by the work at the top level**; the recursive subproblems shrink fast enough (due to the small branching factor and large input reduction) that their total contribution is bounded.



can you analyze and explain this page? you may get some wrong readings because of the handwriting but try to fix and give me the proper explanation with ur own knowledge

Absolutely! Let's break down and analyze the recurrence solution shown in your handwritten page for:

$$T(n) = 3T\left(\frac{n}{4}\right) + \Theta(n^2)$$

◆ TOP SECTION (Recurrence Tree)

This part shows how the recurrence is broken down using the **recurrence tree method**.

✓ Level 0:

- Work done = $c \cdot n^2$ (from the $\Theta(n^2)$ term)

✓ Level 1:

- 3 subproblems, each of size $n/4$
- Each does $c \cdot (n/4)^2 = c \cdot n^2/16$
- Total work:

$$3 \cdot \left(\frac{c \cdot n^2}{16}\right)$$

✓ Level 2:

- $3^2 = 9$ subproblems, each of size $n/16$
- Work per subproblem: $c \cdot (n/16)^2 = c \cdot n^2/256$
- Total work:

$$9 \cdot \left(\frac{c \cdot n^2}{256} \right)$$

✓ General Level i :

- Number of subproblems: 3^i
- Size of each: $\frac{n}{4^i}$
- Work per subproblem:

$$c \cdot \left(\frac{n}{4^i} \right)^2 = c \cdot \frac{n^2}{16^i}$$

- Total work at level i :

$$3^i \cdot c \cdot \frac{n^2}{16^i} = c \cdot n^2 \cdot \left(\frac{3}{16} \right)^i$$

♦ BOTTOM SECTION (Series Expansion)

The handwritten part transitions the recurrence tree into a series:

$$T(n) = c \cdot n^2 \left[1 + \frac{3}{16} + \left(\frac{3}{16} \right)^2 + \dots + \left(\frac{3}{16} \right)^{\log_4 n} \right]$$

This is a **geometric series**:

$$T(n) = c \cdot n^2 \cdot \sum_{i=0}^{\log_4 n} \left(\frac{3}{16} \right)^i$$

♦ Key Observations from the Geometric Series

- Since $\frac{3}{16} < 1$, the geometric series **converges** (bounded by a constant).
- So,

$$\sum_{i=0}^{\log_4 n} \left(\frac{3}{16} \right)^i \leq \sum_{i=0}^{\infty} \left(\frac{3}{16} \right)^i = \frac{1}{1 - 3/16} = \frac{16}{13}$$

Therefore,

$$T(n) = O(n^2)$$

◆ Final Result (Matching the Last Line in the Image):

$$T(n) = \boxed{O(n^2)}$$

The handwritten page follows this logic very well — despite a few unclear spots, the reasoning is sound and mathematically correct.

◆ Extra Note:

If you were to use **Master Theorem** to double-check:

$$T(n) = aT(n/b) + f(n)$$

Here:

- $a = 3$,
- $b = 4$,
- $f(n) = \Theta(n^2)$
- Compare $f(n)$ to $n^{\log_b a} = n^{\log_4 3} \approx n^{0.792}$

Since $f(n) = \Theta(n^2) \gg n^{\log_4 3}$, we are in **Case 3** of the Master Theorem:

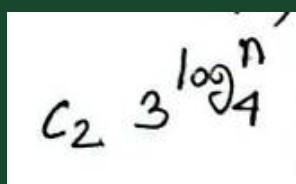
$$T(n) = \Theta(n^2)$$

✓ **Matches the tree method!**

✓ Summary:

- The page correctly applies recurrence tree expansion.
- Each level's total cost is $c \cdot n^2 \cdot \left(\frac{3}{16}\right)^i$
- Total work = geometric sum $\times n^2$
- Final complexity:

$$\boxed{T(n) = \Theta(n^2)}$$



why was this added

Great question.

The term:

$$c_2 \cdot 3^{\log_4 n}$$

was added to account for the **total number of leaf-level subproblems** in the recurrence tree. Let's break this down clearly:

♦ What Happens at the Leaf Level?

At each recursive step:

- The problem is divided into **3** subproblems of size $\frac{n}{4}$, until the size becomes 1 (the base case).
- The depth of the recursion tree is:

$$h = \log_4 n$$

because each time, size reduces by a factor of 4 until it reaches 1.

So at the **last level**, the number of subproblems (i.e. leaves) is:

$$3^{\log_4 n}$$

This represents how many **base case calls** (i.e. $T(1)$) we get at the bottom.

♦ Why Multiply by c_2 ?

Each base case costs **some constant time**, say c_2 , like:

$$T(1) = \Theta(1) = c_2$$

So the **total cost at the leaf level** is:

$$c_2 \cdot 3^{\log_4 n}$$

♦ How Big is That Term?

We can rewrite it using logarithms:

$$3^{\log_4 n} = n^{\log_4 3}$$

So the leaf work is:

$$\Theta(n^{\log_4 3}) \approx \Theta(n^{0.792})$$

This is **less than** $\Theta(n^2)$, so it's not the dominating term — but still part of the total work, so it's included for **completeness** in the summation.

✓ Final Summary:

The term

$$c_2 \cdot 3^{\log_4 n}$$

was added to represent the **total cost of all base-case computations** at the leaves of the recurrence tree.

Although it's not asymptotically dominant (compared to $\Theta(n^2)$), it's part of the full breakdown and ensures the recurrence tree is fully accounted for.